

PROGRAM – 4

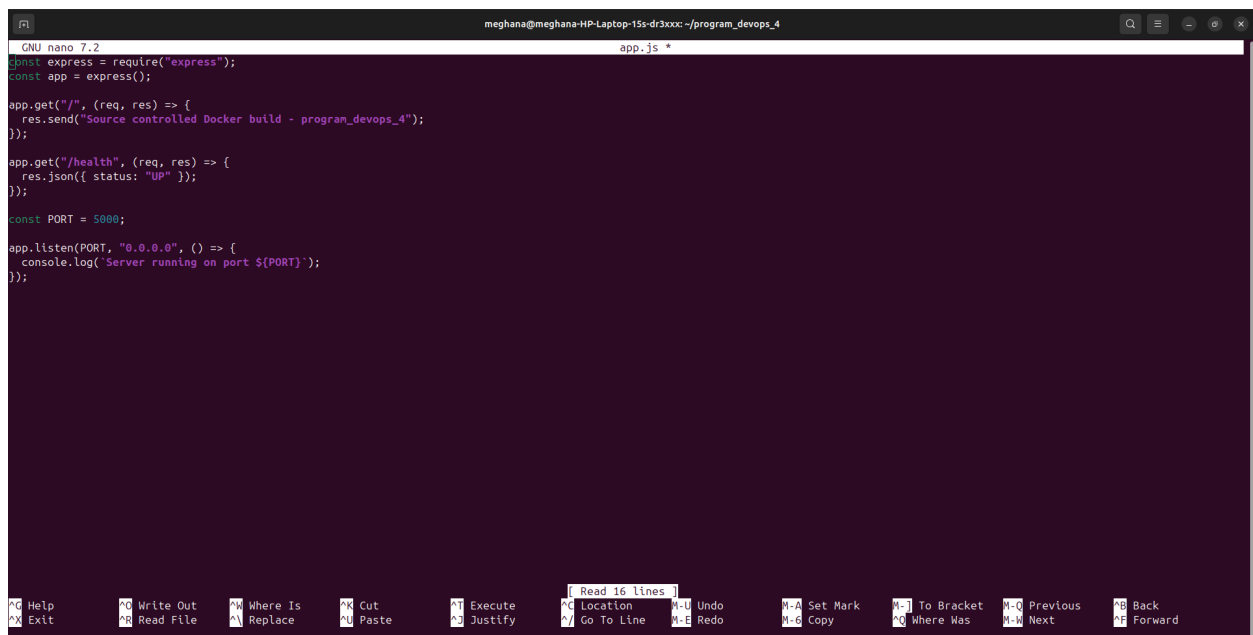
Integrate Git with Docker for Source-Controlled Application Builds

Project Name: program_devops_4

Technology: Git + Docker + Node.js

Step 1: Create project directory

mkdir program_devops_4 cd program_devops_4



```
GNU nano 7.2 app.js *
const express = require("express");
const app = express();

app.get("/", (req, res) => {
  res.send("Source controlled Docker build - program_devops_4");
});

app.get("/health", (req, res) => {
  res.json({ status: "UP" });
});

const PORT = 5000;

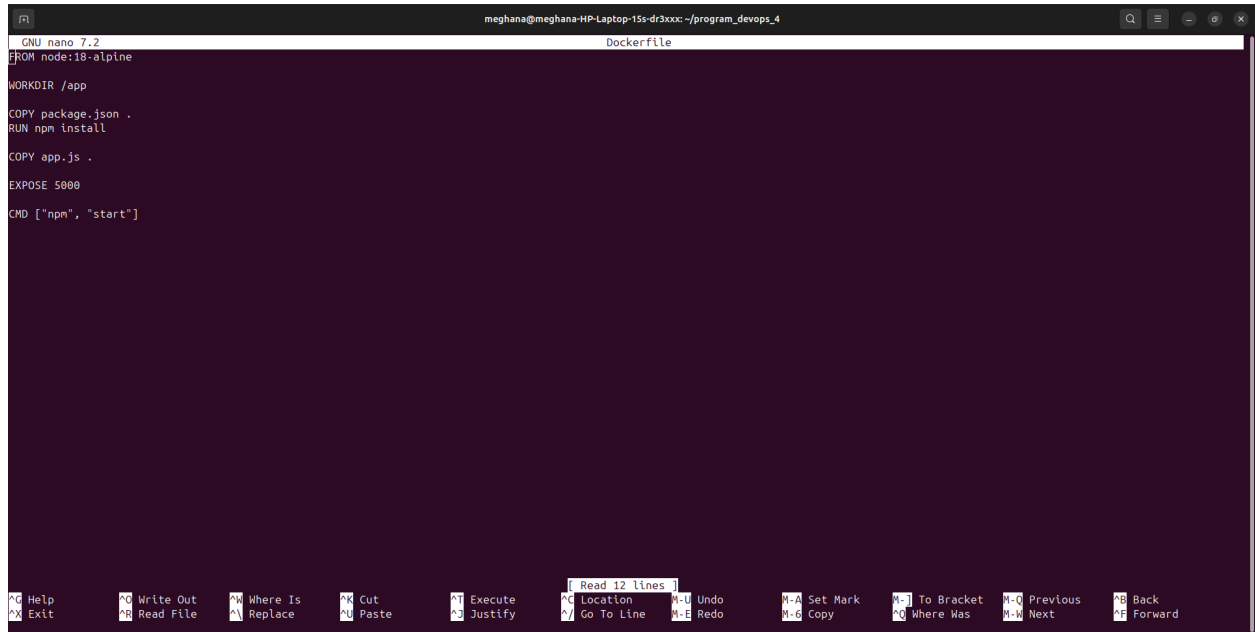
app.listen(PORT, "0.0.0.0", () => {
  console.log(`Server running on port ${PORT}`);
});
```

Step 2: Initialize Git repository

git init

Step 3: Create Node.js application

nano app.js



```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_4
GNU nano 7.2 Dockerfile
FROM node:18-alpine

WORKDIR /app

COPY package.json .
RUN npm install

COPY app.js .

EXPOSE 5000

CMD ["npm", "start"]

[ Read 12 lines ]
[ Location ] [U] Undo [A] Set Mark [A] To Bracket [Q] Previous [B] Back
[X] Exit [R] Read File [W] Where Is [C] Cut [T] Execute [G] Go To Line [E] Redo [G] Copy [Q] Where Was [W] Next [F] Forward
```

Step 4: Create package.json

nano package.json



```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_4
GNU nano 7.2 package.json
{
  "name": "program_devops_4",
  "version": "1.0.0",
  "main": "app.js",
  "scripts": {
    "start": "node app.js"
  },
  "dependencies": {
    "express": "^4.18.2"
  }
}

[ Read 11 lines ]
[ Location ] [U] Undo [A] Set Mark [A] To Bracket [Q] Previous [B] Back
[X] Exit [R] Read File [W] Where Is [C] Cut [T] Execute [G] Go To Line [E] Redo [G] Copy [Q] Where Was [W] Next [F] Forward
```

Step 5: Create Dockerfile

nano Dockerfile

```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_4
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint: git branch -m <name>
Initialized empty Git repository in /home/meghana/program_devops_4/.git/
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$ nano app.js
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$ nano package.json
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$ nano Dockerfile
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$ nano app.js
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$ nano app.js
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$ nano Dockerfile
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$ nano package.json
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$ git add .
git commit -m "Initial commit: Node.js app with Dockerfile"
[master (root-commit) 04f7e2a] Initial commit: Node.js app with Dockerfile
3 files changed, 39 insertions(+)
create mode 100644 Dockerfile
create mode 100644 app.js
create mode 100644 package.json
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$ docker build -t program_devops_4 .
[*] Building 15.6s (11/11) FINISHED
[+] Internal load build definition from Dockerfile
=> transferring dockerfile: 158B
=> [internal] load metadata for docker.io/library/node:18-alpine
=> [auth] library/node:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> transferring context: 2B
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c20fd3ebc498332f249011d118945588d0a15cb9bc4b8ca08d9e
=> [internal] load build context
=> transferring context: 588B
=> CACHED [2/5] WORKDIR /app
=> [3/5] COPY package.json .
=> [4/5] RUN npm install
=> [5/5] COPY app.js .
=> exporting to image
=> exporting layers
=> writing image sha256:74c8380a08ca7d22e27157e7d1c4c391390cdc17eca83cdcb3888d5acfb1f77
=> naming to docker.io/library/program_devops_4
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$
```

Step 6: Add files to Git

git add . git commit -m "Initial commit: Node.js app with Dockerfile"

Step 7: Build Docker image from Git source

docker build -t program_devops_4 .

Step 8: Run Docker container

docker run -p 5000:5000 program_devops_4

Step 9: Commit (if not committed already)

git add . git commit -m "Initial commit for source-controlled Docker build"

Step 10: Add GitHub remote

Step 10: Push code to GitHub

git branch -M main git push -u origin main

```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_4
=> transferring context: 2B
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c20fd3ebc490332f249011d11894558800a35cb9bc4b8ca89d9e
=> [internal] load build context
=> transferring context: 588B
=> CACHED [2/5] WORKDIR /app
=> [3/5] COPY package.json .
=> [4/5] RUN npm install
=> [5/5] COPY app.js .
=> exporting to image
=> exporting layers
=> writing image sha256:74c8380a08ca7d22e27157e7d1c4591390cdc17eca83cdbc38088d5acfb1f77
=> naming to docker.io/library/program_devops_4
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$ docker run -p 5000:5000 program_devops_4
program_devops_4b1.0.0 start
node app.js

Server running on port 5000
^Cnpm error path /app
npm error command failed
npm error signal SIGINT
npm error command sh -c node app.js
npm error A complete log of this run can be found in: /root/.npm/_logs/2026-01-05T06_14_14_000Z-debug-0.log
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$ git remote add origin https://github.com/meghana361/program4_devops.git
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$ git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/meghana361/program4_devops.git'
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$ git branch -M main
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$ git push -u origin main
Username for 'https://github.com': meghana361
Password for 'https://meghana361@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 768 bytes | 768.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/meghana361/program4_devops.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_4$
```

Verify on GitHub:

Open:

https://github.com/meghana361/program_devops_4

meghana361 / program4_devops

Q Type to search

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

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Settings

program4_devops

Public

Pin

Watch 0

Fork 0

Star 0

main 1 Branch 0 Tags

Go to file

Add file

Code

meghana361

Initial commit: Node.js app with Dockerfile

0d77e2a · 5 minutes ago

1 Commit

Dockerfile

Initial commit: Node.js app with Dockerfile

5 minutes ago

app.js

Initial commit: Node.js app with Dockerfile

5 minutes ago

package.json

Initial commit: Node.js app with Dockerfile

5 minutes ago

README

Add a README

Help people interested in this repository understand your project.

Add a README

About

No description, website, or topics provided.

Activity

0 stars

0 watching

0 forks

Releases

No releases published

Create a new release

Packages

No packages published

Publish your first package

Languages

JavaScript 73.6%

Dockerfile 26.4%

Suggested workflows

Based on your tech stack